Higher Colleges of Technology

كليات التقنية العليا

Course Work Project Description and Rubric

Semester	202310	Division	Computer Information Science
Assessment title in Syllabus	Group Project	Program	All Programs - General Studies' Course

Course Code	ICT 2013		
Course Title	Computational Thinking and Coding		
CLOs	CLO4	Accreditation Body	CAA & CIPS
Course Instructor		CRN	
Assessment Weight	25%	Submission Date	Week 13 (Specify exact date here)

For Group Work submissions an additional individual assessment will be conducted.

Grades for the students in one group will vary based on the individual performance in the additional assessment.

Student Declaration:

Academic Integrity Statement

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

إفادة النزاهة الأكاديمية

وفقًا لسياسة كليات التقنية العليا للنزاهة الأكايمية

- على الطلبة الالتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبيّن وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- **في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكايمية، سيتعرض الى العقوبات الموضحة في السياسات ذات الصلة.**

This assignment is entirely my own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

Student (s) Signature: ______________________

____

Student Name(s):	*Maryam Ali*	*Nourah Alghfeli*	*Hessa Khalfan*
Student HCT ID(s):	H00535836	H00535834	H00535166

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Section No.	1	2-a	2-b	Total	%
Marks Allocated					
Marks Obtained					

Chatbot (e.g. ChatGPT) statement

This submission does not allow you to use any AI tool to complete. Any breach of this requirement will be treated as a plagiarism attempt and shall be subject to the HCT Academic Integrity Policy.

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Note to students- Please read and then delete/replace the writing in yellow and complete all parts.

PROJECT INTRODUCTION

Describe all the essential information related to the background of your proposed Data Analysis. Minimum 100 words

- Description of the purpose of the project – Aim of the project.

This information refers to the number of births for both genders (Female and Male) and their region in the United Arab Emirates from 1997 to 2015. Our data has six columns (Year, Indecator_Ar, Indecator_En, Type_Ar, Type_En and Value) and 133 rows. Each row is a different year and columns give details about the babies. Our objective is to evaluate this data and determine which years had an increase in births and which year saw a drop in each of the seven emirates. This information was obtained from Bayanati (https://bayanat.ae/en/Data), the UAE's official open data site under theme of Health, Vitality, and Reproductive Health.

ANALYSIS OBJECTIVES

List the questions (10 questions) of the Project. These questions should be relevant, by providing useful information that may help in taking decisions.

Refer to the Project brief for the types of questions.

Q1: What is the total number of births in UAE from 1997 to 2015?

Q2: What years had a birth value of more than 700 in Umm Al-Quwain?

Q3: How many babies were born in" Ras Al-Khaimah" or "Fujairah" from 1997 to 2015?

Q4: How many people were born in each emirate over these years?

Q5: What are the top five years in terms of birth value?

Q6: In 2004, What is the average number of births in each emirate?

Q7: What is the average number of births per year in the UAE from 1997 to 2015, sorted by year in ascending order?

Q8: Is the number of births increasing or decreasing?

Q9: Compare the number of births in each emirate.

Q10: What is the birth rate in each emirate in relation to the Emirates?

DATA ACQUISITION AND CLEANING

Code to read the data from Excel / CSV / HTML.

Code to clean the unnecessary data, by removing, replace the missing data and renaming the columns.

Explanation of why data clean needed (for your data).

We initiated our Python code by employing the statement (import pandas as pd) , introducing the Pandas library as a crucial tool for reading and manipulating our database.

We read the data from Excel and store in the data frames name live read from excel file, and the names of the target sheet is Live Births Emirate.

-import pandas as pd

-live = pd.read_excel("live-births-emirate1997-2015.xlsx", sheet_name ="Live Births Emirate")

Import pandas

```
import pandas as pd
```

Read Data

```
[81] live = pd.read_excel("live-births-emirate1997-2015.xlsx", sheet_name ="Live Births Emirate", skiprows=0)
```

```
live.shape
```

```
(133, 6)
```

Using dropna() to remove the columns that are ,missing all its data or currently not needed

Using fillna to Fill in missing data with zeros.

-live.drop(columns = ["Indecator_Ar","Type_AR"], inplace= True)

-live.dropna(how='any', inplace=True)

-live.fillna(0,inplace=True)

-live.rename(columns = {"Indecator_En":"Indecator","Type_EN":"Emirate", "Value":"Births"}, inplace = True) cleaning

```
[7] #We will eliminate the columns that are currently not required.

    live.drop(columns = ["Indecator_Ar","Type_AR"], inplace= True)
```

```
[8] #We will delet rows that are missing all its data permanently
    live.dropna(how='any', inplace=True)
```

```
[9] #We will replace any missing data with zero (0)
    live.fillna(0,inplace=True)
```

```
#We will rename the data to make it easier to identify.

live.rename(columns = {"Indecator_En":"Indecator","Type_EN":"Emirate", "Value":"Births"}, inplace = True)
```

```
live.shape

(133, 6)
```

DATA AND EXPLORATORY ANALYSIS

Screen Shot of the code and its output with explanation of how your data analysis meets the project objectives and produced the required findings as per the project specification.

This part of analysis must include – Statistics, Filter, Calculations, Group, Sort.

Screen Shot of the code and its output with short explanation about the data, sample rows, columns, total rows, and descriptions. This part is to get familiar with your data and summarize their main characteristics

statistical methods (sum)

Q1: What is the total number of births in UAE from 1997 to 2015?

```
[ ] #We will use statical methods to (sum()) to find total number of births
    live["Births"].sum()

    1316631
```

```
#We will use statical methods to (sum())
to find total number of births
live["Births"].sum()
```

conditional filtering with more than one condition (and)

Q2)What years had a birth value of more than 700 in Umm Al-Quwain?

```
#We used this code to filter our data and display the years were the births in UAQ are more than 700
live[(live["Emirate"] == "Umm Al-Quwain") & (live["Births"]>700)]
```

	Year	Indecator	Emirate	Births
36	1997	Live Births by Emirate	Umm Al-Quwain	761
38	1999	Live Births by Emirate	Umm Al-Quwain	722
39	2000	Live Births by Emirate	Umm Al-Quwain	716
40	2001	Live Births by Emirate	Umm Al-Quwain	709
42	2003	Live Births by Emirate	Umm Al-Quwain	756
43	2004	Live Births by Emirate	Umm Al-Quwain	702
44	2005	Live Births by Emirate	Umm Al-Quwain	727
47	2008	Live Births by Emirate	Umm Al-Quwain	758
105	2009	Live Births by Emirate	Umm Al-Quwain	836
110	2014	Live Births by Emirate	Umm Al-Quwain	889
111	2015	Live Births by Emirate	Umm Al-Quwain	1228

```
live[(live["Emirate"] ==
"Umm Al-Quwain") &
(live["Births"]>700)]
```

conditional filtering with more than one condition (or)

Q3:How many babies where born in "Ras Al-Khaimah" or "Fujairah" from 1997 to 2015

```
#We used this code to display all births in RAK or Fujairah

live[(live["Emirate"]=="Ras Al-Khaimah")|(live["Emirate"]=="Fujairah")]
```

	Year	Indecator	Emirate	Births
125	2015	Live Births by Emirate	Ras Al-Khaimah	4557
120	2010	Live Births by Emirate	Ras Al-Khaimah	4281
122	2012	Live Births by Emirate	Ras Al-Khaimah	4139
124	2014	Live Births by Emirate	Ras Al-Khaimah	4073
97	2015	Live Births by Emirate	Fujairah	4033
119	2009	Live Births by Emirate	Ras Al-Khaimah	4032
123	2013	Live Births by Emirate	Ras Al-Khaimah	4004
121	2011	Live Births by Emirate	Ras Al-Khaimah	3999
96	2014	Live Births by Emirate	Fujairah	3894
95	2013	Live Births by Emirate	Fujairah	3892
94	2012	Live Births by Emirate	Fujairah	3822
71	2008	Live Births by Emirate	Ras Al-Khaimah	3625
92	2010	Live Births by Emirate	Fujairah	3591
93	2011	Live Births by Emirate	Fujairah	3548
67	2004	Live Births by Emirate	Ras Al-Khaimah	3450
66	2003	Live Births by Emirate	Ras Al-Khaimah	3361
65	2002	Live Births by Emirate	Ras Al-Khaimah	3329
64	2001	Live Births by Emirate	Ras Al-Khaimah	3258
68	2005	Live Births by Emirate	Ras Al-Khaimah	3230
70	2007	Live Births by Emirate	Ras Al-Khaimah	3214
91	2009	Live Births by Emirate	Fujairah	3202
63	2000	Live Births by Emirate	Ras Al-Khaimah	3096
69	2006	Live Births by Emirate	Ras Al-Khaimah	3013
61	1998	Live Births by Emirate	Ras Al-Khaimah	2957
62	1999	Live Births by Emirate	Ras Al-Khaimah	2941
23	2008	Live Births by Emirate	Fujairah	2861
60	1997	Live Births by Emirate	Ras Al-Khaimah	2755
22	2007	Live Births by Emirate	Fujairah	2625
18	2003	Live Births by Emirate	Fujairah	2396
19	2004	Live Births by Emirate	Fujairah	2389
20	2005	Live Births by Emirate	Fujairah	2319
21	2006	Live Births by Emirate	Fujairah	2311
12	1997	Live Births by Emirate	Fujairah	2293
16	2001	Live Births by Emirate	Fujairah	2176
15	2000	Live Births by Emirate	Fujairah	2175
14	1999	Live Births by Emirate	Fujairah	2147
17	2002	Live Births by Emirate	Fujairah	2094
13	1998	Live Births by Emirate	Fujairah	2052

```
#We used this code to display all births in RAK and Fujairah

live[(live["Emirate"]=="Ras Al-Khaimah")|(live["Emirate"]=="Fujairah")]
```

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Data Analysis using group the data by column(s)

Q4:How many people were born in each emirate over these years?

```
#This code counts the total number of births for each Emirate.
live.groupby('Emirate')[['Births']].sum()
```

Emirate	Births
Abu Dhabi	530880
Ajman	81101
Dubai	392606
Fujairah	53820
Ras Al-Khaimah	67314
Sharjah	177009
Umm Al-Quwain	13901

```
live.groupby('Emirate')[['Births']].sum()
```

Data Analysis using sorting the data in ascending/ descending order

Q5:What are the TOP 5 years in birth value?

```
#We have to reorder the data according to there birth value (descending order big->small)
#Then we ask it to print the first 5 (Biggest 5)
live.sort_values(["Births"], ascending=False, inplace=True)
live.head(5)
```

	Year	Indecator	Emirate	Births
104	2015	Live Births by Emirate	Abu Dhabi	38818
103	2014	Live Births by Emirate	Abu Dhabi	38035
102	2013	Live Births by Emirate	Abu Dhabi	35945
101	2012	Live Births by Emirate	Abu Dhabi	34358
100	2011	Live Births by Emirate	Abu Dhabi	32084

```
datafor2004 = live[live["Year"] == 2004].groupby(['Year','Emirate'])[['Births']].mean()
datafor2004.sort_values(['Births'], ascending=False)
```

Data Analysis using combination of sorting, condition filter and/or grouping.

Q6: How many individuals were born in each emirate in 2004?

```
#It shows the average number of births in 2004 for each combination of "Year" and "Emirate."
#then sorted in decreasing order by the mean number of births.

datafor2004 = live[live["Year"] == 2004].groupby(['Year','Emirate'])[['Births']].mean()
datafor2004.sort_values(['Births'], ascending=False)
```

Year	Emirate	Births
2004	Abu Dhabi	26261.0
	Dubai	17013.0
	Sharjah	10243.0
	Ras Al-Khaimah	3450.0
	Ajman	3055.0
	Fujairah	2389.0
	Umm Al-Quwain	702.0


```
#It shows the average number of births in 2004 for each combination of "Year" and
"Emirate."
#then sorted in decreasing order by the mean number of births.

datafor2004 = live[live["Year"] == 2004].groupby(['Year','Emirate'])[['Births']].mean()
datafor2004.sort_values(['Births'], ascending=False)
```

Data Analysis using combination of sorting, condition filter and/or grouping.

Q7:What is the average number of births per year in the UAE from 1997 to 2015, sorted by year in ascending order?

```
#this code filters the original DataFrame for the years between 1997 and 2015,
#calculates the average number of births for each year
#resets the index, and then sorts the result by the 'Year' column in ascending order.


dataforyears = live[(live['Year'] >= 1997) & (live['Year'] <= 2015)]
average_births_per_year = dataforyears.groupby('Year')['Births'].mean().reset_index()
average_births_per_year.sort_values('Year', ascending=True)
```

	Year	Births
0	1997	6622.857143
1	1998	6876.571429
2	1999	7094.142857
3	2000	7669.428571
4	2001	8019.428571
5	2002	8295.714286
6	2003	8737.857143
7	2004	9016.142857
8	2005	9231.857143
9	2006	8995.571429
10	2007	9669.857143
11	2008	9825.571429
12	2009	10909.428571
13	2010	11375.000000
14	2011	11992.857143
15	2012	12796.857143
16	2013	13362.714286
17	2014	13694.285714
18	2015	13904.000000


```
dataforyears = live[(live['Year'] >= 1997) & (live['Year'] <= 2015)]
average_births_per_year =
dataforyears.groupby('Year')['Births'].mean().reset_index()
average_births_per_year.sort_values('Year', ascending=True)
```


DATA ANALYSIS - VISUALIZATION

Screen Shot of the code and its output with explanation of how your findings can be visualized.

This part of analysis must include – Code and output of the Plot Graph(s) with appropriate color, title, x-axis, and y-axis, with explanation of the findings. Use different types of appropriate charts to meet your objectives.

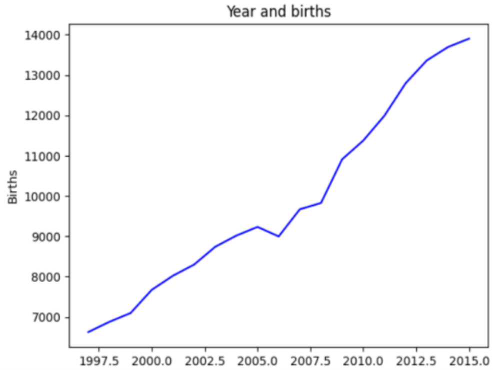
process to do the data analysis

Q8: Is the number of births increasing or decreasing?

```
[21] # In this code we used the variable "Q8" to save the rest of the code in it, then we used the command "groupby" to group elements together
#with the addition of the "mean()" function to calculate the average of the set of numbers.
#Then we used the "plot.line" function to creat a line chart,then we chose a sutable title and blue color for the line, then we set a name for the x,y label.
Q8 = live.groupby(["Year"])["Births"].mean()
line = Q8.plot.line(title = "Year and births" , color="blue")

line.set_xlabel("Year")
line.set_ylabel("Births")
```

```
Text(0, 0.5, 'Births')
```




```
Q8 = live.groupby(["Year"])["Births"].mean()
line = Q8.plot.line(title = "Year and births" ,
color="blue")

line.set_xlabel("Year")
line.set_ylabel("Births")
```

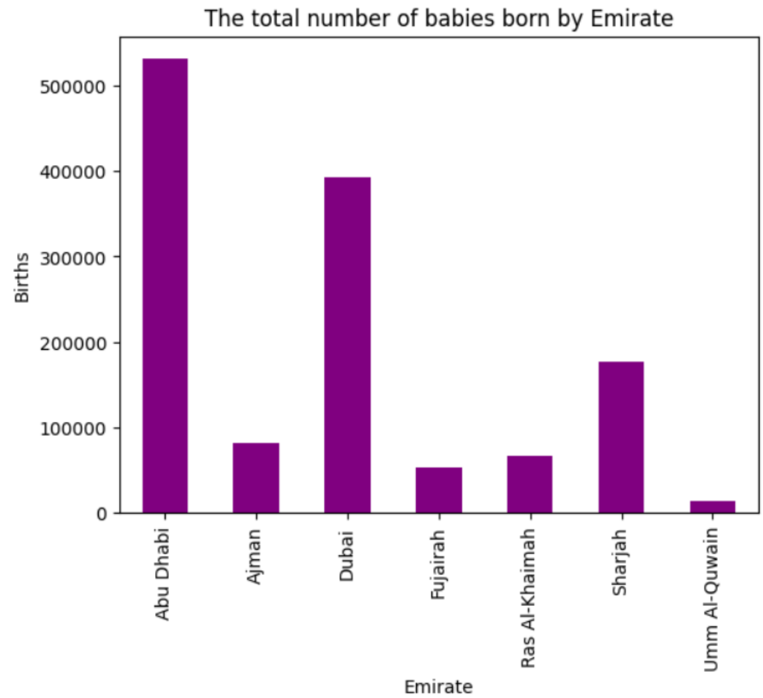
Q9: Compare the number of births in each Emirate

```
[19] # In this code we used the variable "Q9" to save the rest of the code in it, then we used the command "groupby" to group elements together
#with the addition of the "sum()" function to calculate the sum of a collection of numbers.
#Then we used the "plot.bar" function to creat a bar chart,then we chose a sutable title and purple color for the bar, then we set a name for the x,y label.

Q9 = live.groupby(["Emirate"])["Births"].sum()
bar = Q9.plot.bar(title = "The total number of babies born by Emirate",color="purple")

bar.set_xlabel("Emirate")
bar.set_ylabel("Births")
```

```
Text(0, 0.5, 'Births')
```



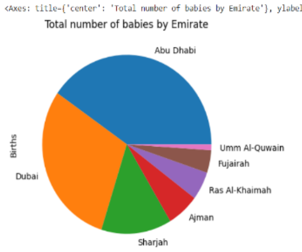

```
Q9 = live.groupby(["Emirate"])["Births"].sum()
bar = Q9.plot.bar(title = "The total number of babies born by
Emirate",color="purple")

bar.set_xlabel("Emirate")
bar.set_ylabel("Births")
```

Q10: What is the birth rate in each Emirate in relation to the Emirates?

```
[20] # In this code we used the variable "Q10" to save the rest of the code in it, then we used the command "groupby" to group elements together
     #with the addition of the "sum()" function to calculate the sum of a collection of numbers. And we used the "sort_values" to arrange the data.
     #Then we used the "plot.pie" function to creat a pie chart,then we chose a sutable title for the pie chart.
     Q10 = live.groupby(["Emirate"])["Births"].sum()
     Q10 = Q10.sort_values(ascending = False)
     Q10.plot.pie(title = "Total number of babies by Emirate")

<Axes: title={'center': 'Total number of babies by Emirate'}, ylabel='Births'>
```




```
Q10 = live.groupby(["Emirate"])["Births"].sum()

Q10 = Q10.sort_values(ascending = False)
Q10.plot.pie(title = "Total number of babies by Emirate")
```

EXECUTIVE SUMMARY

Write a executive summary for minimum 200 words that summarize your findings and include what are the pro and cons of the data / system. Write your recommendations that could benefit / improve the process / system.

Pros Recommendation Cons

The data provides a comprehensive overview of the birth distribution across the Emirates, enabling targeted research in every region. Makes it possible to compare the birth rates of several Emirates. The information highlights patterns and variances spanning almost 20 years, providing insightful information about the temporal evolution of birth rates. When examining changes over time, it assists determine if the number of births is rising or falling over time. helpful for comparing several categories; in this instance, it let us see how the birth rates in the Emirates differed from one another.

The total number of births that we were previously aware of was 1316631. The birth statistics for the Emirates from the dataset that is supplied shows notable regional differences in the number of births. Abu Dhabi and Dubai are the most populous Emirates, with 530,880 and 392,606 births, respectively, over this period. There are also a lot of births in Ajman, Ras Al-Khaimah, and Sharjah, but not as many in Fujairah or Umm Al-Quwain. With this information, authorities may plan social services, healthcare, and education according to participant needs and numbers. It facilitates the process of identifying the regions that most need increased healthcare services and attention. To give a more thorough picture of the observed patterns, supplement the dataset with additional contextual variables such changes in the socioeconomic landscape, population transitions, or policy changes.

The dataset from the temporal analysis of birth rates from 1997 to 2015 shows an increase that is evolving with time. The average yearly birth rate increased until 2015, when it reached its highest point. The line graph shows that although the number of births increases each year, it does show a decline in the years 2006 and 2008. While the overall trend is evident, the dataset lacks information on specific causes or environmental factors that might influence variations.

REFERENCES

Include all the external references that you might have used. APA / MLA referencing style may be used. If you're not aware of MLA or not fully confident in using it, refer to the library resources including librarians.

Data Source:

https://bayanat.ae/en/Data

Data Link:

https://admin.bayanat.ae/Home/DatasetInfo?dID=vMeX8LEWLHbeQ2cBmwzQuQw22lDMVNo9d45dlCDcCVY

Google Colab:

https://colab.research.google.com/drive/1QRhAHx_JN6RvTkibvgQq4CW63rjralt9?usp=sharing#scrollTo=0RpZq1WD77t_